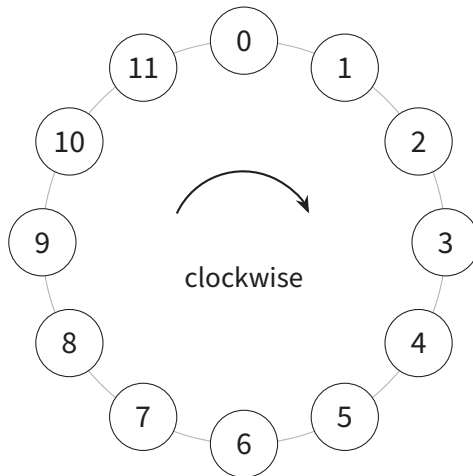


The twelve-place code

Name: _____

Date: _____

Use positions 0 through 11 clockwise. A shift $+k$ moves every input k places clockwise, wrapping after 11. Decoding reverses that shift. A turn repeats the **same** shift.



1. A broken message?

Input 9 became output 1. Find the shift $+k$, where k is between 0 and 11. Check whether input 2 could become 6 under the same shift. Decode the output sequence **1, 4, 10**.

My key, check, and decoded message:

2. Follow one input.

Starting at 0, repeat that shift until the first return. Then try starting at 1. Find **all** its separate loops, including every position exactly once.

All loops, and the first return time:

Try it with objects. Explain by pointing, talking, or drawing.

Classify every shift

3. Find a rule, not just a table.

Try enough shifts to spot a rule: start at 0, find the first positive return, and count separate loops. Use clockwise/counterclockwise symmetry. The full table is optional; move to page 3 when ready to explain.

Shift	Turns to return	Number of loops	Visits all 12?
0			
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			

4. Predict before you test.

On an 18-place ring, predict the return times for +6 and +5. What number connects the ring size, the step size, and the number of loops?

My conjecture and evidence:

Try it with objects. Explain by pointing, talking, or drawing.

Why the rule works

5. Count distance around the ring.

After t turns of $+k$ on an n -place ring, you have moved kt spaces. You are back at the start exactly when kt is a whole multiple of n . Use this to explain $+4$ on a 12-ring.

The first return and why none earlier works:

6. Shrink the common factor.

For $+8$ on 12 places, return means 12 divides $8t$, or 3 divides $2t$. Draw three equal t -strips: together they total $3t$. If two strips total a multiple of 3, why does removing them leave t a multiple of 3? Explain the first return.

My strip argument; optional general rule:

7. Undoing is different from visiting.

The $+4$ rule does not visit every place from 0. Does it still give a reversible code on *all* twelve inputs? Find its inverse. Then compare $+4$ followed by $+7$ with $+7$ followed by $+4$. Explain for every starting place.

Inverse, combined shift, and why the order agrees:

Optional, with an adult: Use the guide's subtraction-of-strips argument to justify coprime cancellation, then prove the gcd rule. Until then, label the general rule a conjecture or a supplied theorem.

Try it with objects. Explain by pointing, talking, or drawing.